

What is Transforming the Smart Grid

Smart grid is the paradigm for a class of technology being designed to join forces and modernise our entire energy industry. This story gives a glimpse of how it is being done and what it means for you

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Imagine a country with no load shedding, power cuts or black-outs. A new era of reliable and efficient electricity supply is what the smart grid concept promises.

Integrating modern digital electronics technology into the electrical power grid enables the utility to detect, monitor and respond digitally to any changes, thus making it a smarter version of the grid and earning it the name 'smart grid' in the process.

The current power grid is based on decades-old technology that makes it prone to a lot of outages. An imbalance in the stress on a power grid could cause electricity disruption. If this imbalance is not corrected in time, it could have a domino effect, affecting the functioning of other critical utilities.

By implementing the smart grid concept, the grid will be armed with a plethora of sensors, controllers, communication equipment and new technology that enables the grid to be truly automated—thus enabling it to promptly respond to any incidents in the grid. "This also means that the smart grid technology is a multidisciplinary area, requiring power engineers, IT, telecom and modelling forecasting to work together," says Sandra Diethelm, CEO, Exuta (India) Pvt Ltd.

How a smart grid responds to an outage

When a disturbance in the grid is initiated, the sensing and monitoring mechanism of the grid detects the cause of the outage and enables the



Elements of a smart grid

grid to take measures to quarantine it so it does not affect the rest of the grid. In the quarantined grid, the technology begins working on recovering electricity distribution to the critical services first—providing power to hospitals, police institutions, communications, etc. It will also loop in consumer-controlled power-generation sources when the utility is unable to provide enough power, thus fostering energy trade within the community as well.

Evolution of smart grid technologies

Here we look at some recent advances in various aspects of smart grid technologies.

Even smarter meters. Smart meters are not only limited to measurement, but also play an integral part in deriving the quality of power. "An energy audit meter not only measures the power on a relative parameter, but also measures its quality," explains A.V. Srinivasan, area manager—south, M.B. Control and

"Smart grid is not always about smart meters. Australia's smart meter rollout made sense in Melbourne but not anywhere else. A national smart meter rollout does not appear to be on the agenda today, nor probably in the near future."

—Sandra Diethelm, CEO, Exuta (India) Pvt Ltd

Systems Pvt Ltd. He further explains that smart audit meters analyse the power by decoding parameters such as harmonics, power factor and so on.

The newer meters not only monitor and provide data to services connected to it but also provide control over the network to designated control centres.